Question 17

Q: An image captured under uneven lighting shows poor contrast. Explain how the image formation process contributes to this issue and suggest corrective measures.

Theoretical Concept: Uneven Lighting and Contrast

Uneven lighting occurs when the illumination component *i(x,y)* of the image formation model varies heavily across the scene (e.g., shadows on one side). This compresses the dynamic range of the reflectance in the dark areas, causing poor contrast.

Corrective Measures: Local/Adaptive Histogram Equalization (CLAHE) is perfect here. It divides the image into tiles and enhances contrast locally, preventing over-amplification of noise.

💻 OpenCV Python Solution

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```
import cv2  # Import OpenCV for image processing algorithms
import numpy as np                                          # Import Numpy for mathematical matrix operations
import matplotlib.pyplot as plt                             # Import Matplotlib to visualize the output on screen

# Step 1: Create a synthetic image with an object hidden in a shadow gradient
img = np.zeros((100, 200), dtype=np.uint8)  # Create a blank black image array filled with zeros
cv2.circle(img, (100, 50), 30, 150, -1)  # Draw a filled or outlined circle on the image
gradient = np.tile(np.linspace(0.1, 1.0, 200), (100, 1))  # Repeat an array to build a larger image or gradient
```

```
shadow_img = (img * gradient).astype(np.uint8)

# Step 2: Create a CLAHE object (Contrast Limited Adaptive Histogram Equalization)
clahe = cv2.createCLAHE(clipLimit=2.0, tileGridSize=(8,8))  # Create CLAHE object for adaptive contrast enhancement

# Step 3: Apply CLAHE to locally fix the contrast across the gradient
fixed_img = clahe.apply(shadow_img)  # Apply CLAHE to locally enhance image contrast

# Step 4: Visualize the correction
plt.figure(figsize=(8,4))  # Create a new figure window for display
plt.subplot(121), plt.imshow(shadow_img, cmap='gray'), plt.title('Uneven Contrast')  # Place this image in a subplo
plt.subplot(122), plt.imshow(fixed_img, cmap='gray'), plt.title('CLAHE Restored')  # Place this image in a subplot
plt.show()  # Show the final plot window to the user
```

🎯 Expected Output

A side-by-side plot where a dim circle lost in a shadow is dramatically brought into sharp, normalized focus without blowing out the highlights on the bright side of the image.

🚀 How to Execute & Submit

1. Google Colab Execution:

- Open [Google Colab](https://colab.research.google.com) and create a New Notebook.
- Click the 📋 **Copy** button on the code block above.

- Paste it into a Colab cell and press **Shift + Enter** to run.

2. Uploading to GitHub:

- In Colab, go to **File** > **Save a copy in GitHub**.
- Authorize your GitHub account.
- Select your Computer Vision Lab repository and click **OK** to commit the code.